Homework Assignment 1

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Rutgers University

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Problem 1:

(a)

$$\text{P(At least 1 pair)} 1 - \frac{\binom{10}{6} * 2^6}{\binom{20}{6}} \approx 0.6533$$

(b)

$$\text{P(Exactly 2 pair)} = \frac{\binom{10}{2}\binom{8}{2}2^2}{\binom{20}{6}}$$

Then,

$$\begin{aligned}\text{P(Exactly 2 pair} \mid \text{At least 1 pair)} &= \frac{\text{P(Exactly 2 pair} \cap \text{At least 1 pair)}}{\text{P(At least 1 pair)}} \\ &= \frac{5040}{25340} \\ &\approx 0.1991\end{aligned}$$

Problem 2: We can use total probability, either the second card is heart suited or it isn't

$$\begin{aligned}P(H_1) &= P(H_1 \mid H_2 \cap H_3 \cap H_4) * P(H_2|H_3 \cap H_4) + P(H_1 \mid H_2^c \cap H_3 \cap H_4) * P(H_2^c|H_3 \cap H_4) \\ &= \frac{10}{49} * \frac{11}{50} + \frac{11}{49} * \frac{39}{50} \\ &= 0.22\end{aligned}$$

Problem 3:

The only chance for HHHH to happen is at the very start since for it to happen after that, T must precede it. Therefore, with the law of total probability, we conclude

$$\begin{aligned}\text{P(THHH before HHHH)} &= 1 - P(HHHH) \\ &= 1 - p^4\end{aligned}$$

Problem 4:

(a) Let A = Suspect A is the burglar, B = Suspect B is the burglar, G = Glove is at scene

$$\begin{aligned} P(A \mid G) &= \frac{P(G \mid A)P(A)}{P(G \mid A)P(A) + P(G \mid B)P(B)} \\ &= \frac{1 * 0.5}{1 * 0.5 + 0.5 * 0.1} \\ &\approx 0.909 \end{aligned}$$

(b) Let O = Suspect B owns the glov

$$\begin{aligned} P(O \mid G) &= \frac{P(G \mid O)P(O)}{P(G)} \\ &= \frac{1 * 0.1}{0.55} \\ &\approx 0.182 \end{aligned}$$

Problem 5:

$$\begin{aligned} &P(x = 0) = P(N_1 < N_2) = 1/2 \\ &P(x = 1) = P(N_1 > N_2) * P(N_1 < N_3) = 1/2 * 1/3 = 1/6 \\ &P(x = 2) = P(N_1 > N_2, N_3) * P(N_1 < N_4) = 1/3 * 1/4 = 1/12 \\ &P(x = 3) = (3!)/5! = 1/20 \\ &P(x = 4) = P(N_1 = 5) = 1/5 \end{aligned}$$